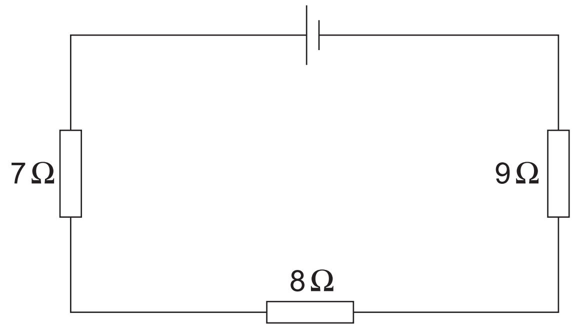
Examiner only

3. (a) The diagram shows three resistors connected in series.



(i) Use an equation from page 2 to calculate the total resistance of the circuit. [1]

total resistance = ... Ω

(ii) The battery voltage is 12 V.
Use an equation from page 2 to calculate the current in the circuit. [2]

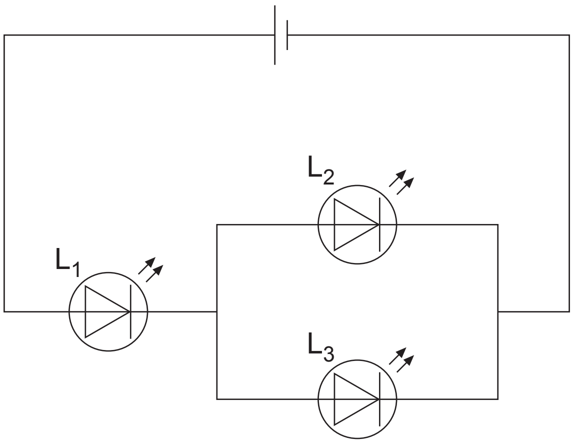
current = ... A

(iii) Use an equation from page 2 to calculate the power produced by the battery. [2]

power = ... W

Examiner only

(b) The circuit below contains three **identical** components, L_1, L_2 and L_3, connected to a battery.



(i) Tick (✓) the box next to the name of these components. [1]

Lamp ☐

LED ☐

LDR ☐

(ii) The current in L_2 is 8 mA.

I. Write down the current in L_3. current = ... mA [1]

II. Write down the current in L_1. current = ... mA [1]

(iii) I. State which **two** components (L_1, L_2 or L_3) have the same voltage across them. [1]

.... and ....

II. State which **two** components (L_1, L_2 or L_3) have voltages that add up to the battery voltage. [1]

.... and ....

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